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Public R&D Investments and Private-sector Patenting: Evidence from NIH Funding Rules

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1 Big picture

- **Question.** Does publicly funded biomedical research at the NIH spur follow-on innovation by private pharmaceutical and biotechnology firms?
- **Core challenge.** Public science is supposed to generate diffuse, non-rival spillovers, so the key empirical problems are: how to link specific research expenditures to downstream commercial outcomes, how to deal with endogenous targeting of funding, and how to measure net effects after possible crowd-out of private R&D.
- **Setting.** The paper studies U.S. NIH grant funding from 1980 to 2005 and links it to life-science patents granted between 1980 and 2012.
- **Unit of analysis.** The main unit is a *disease-science-time* cell, abbreviated DST: a disease area defined by the funding Institute, a science area defined by the reviewing study section, and a year.
- **Main empirical contribution.** The authors build a bibliometric bridge from grants to publications to patents, and they allow the data to reveal spillovers across diseases and across time rather than imposing fixed disease-specific lags.
- **Main identification contribution.** They exploit rigidities in NIH peer review rules. Grants are scored by science-based study sections but funded by disease-based Institutes, which creates quasi-random “windfall” funding variation across DSTs near Institute-specific funding cutoffs.
- **Headline result.** Their preferred IV estimate implies that an additional \$10 million in NIH funding generates about 2.7 extra private-sector patents in related research areas. This is roughly one patent for every two to three NIH grants.
- **Crowd-out result.** The effect on *total related private-sector patents* is positive, so the data do not support full crowd-out of private innovative effort.
- **Spillover result.** More than half of the patents induced by NIH funding occur in disease areas different from the original disease area that funded the research.
- **Valuation result.** Depending on how one values drug-related patents, a \$10 million increase in NIH funding is associated with roughly \$14–\$24 million in downstream drug sales, and one market-value calculation gives about \$30 million in firm value.
- **Why this matters.** The paper gives rare causal evidence on the private innovation returns to public science, and it shows that disease-targeted biomedical funding creates broad cross-disease knowledge spillovers.

2 Data and measurement (key objects)

2.1 Institutional background: how the NIH allocates funds

- The NIH funded about 28% of U.S. medical research in 2008, compared with 37% by pharmaceutical firms, 15% by biotechnology firms, and 7% by medical device firms.
- The agency is organized into Institutes or Centers (ICs), typically around disease or body-system areas. Each IC receives its own congressional appropriation.
- Scientific review is carried out by standing review committees called *study sections*. Study sections are organized around scientific methods or topics rather than diseases.

- In the paper’s logic, the Institute identifies the *disease* margin and the study section identifies the *science* margin.
- Applications receive a **raw score** from reviewers, then a within-study-section normalized **science rank**. Institutes fund applications in order of this science rank until the Institute budget is exhausted. The worst funded score is the Institute’s *payline*.
- This produces the crucial institutional wedge in the paper: grants are *evaluated* within science areas but *funded* within disease budgets.

2.2 Main data sources

- **NIH administrative data.** The authors use IMPAC II grant-application data with grant assignments, scores, study sections, Institutes, and funding outcomes.
- **Publication data.** PubMed provides grant-publication acknowledgment information, which lets the authors link grants to the papers they funded.
- **Patent data.** USPTO life-science patents are linked to papers through patent-to-publication citations.
- **Drug-linked patent data.** The FDA Orange Book and IMS Patent Focus are used to identify patents associated with approved drugs and advanced drug candidates.

2.3 Sample construction

- The grant sample contains **153,076** funded NIH grants from 1980 to 2005 that were reviewed in chartered study sections.
- These grants span **17** Institutes and **624** study sections, aggregating to **14,085** DSTs.
- The patent sample contains **315,982** life-science patents granted between 1980 and 2012.
- Of these patents, **232,276** are classified as private-sector patents and **83,394** as public-sector patents.

2.4 Grant-level variables and grant composition

- Average grant funding is about **\$1.56 million** in 2010 dollars.
- About **73.7%** of grants are R01-equivalent project grants.
- About **59.5%** are new grants rather than renewals.
- A grant is linked to an *acknowledged publication* if a paper reports that grant’s project number.
- A grant is directly linked to a patent when a patent cites a paper that acknowledges that grant.

2.5 Two patent outcome measures

2.5.1 1. Patents citing NIH-funded research

- The first outcome uses a direct grant → publication → patent chain.
- Define a patent as citation-linked to a DST if it cites a publication that acknowledges funding from that DST.

- This measure is flexible in *space* and *time*: the patent may be in a different disease area, and the lag between grant and patent is not imposed by the researcher.
- This outcome captures the patents that explicitly build on NIH-funded research, but it misses indirect diffusion channels and cannot separately identify whether the patent would have existed without public funding.

2.5.2 2. Patents related to NIH-funded research

- The second outcome defines a broader intellectual neighborhood around each DST.
- The authors match grant-funded publications to *similar* publications using the PubMed Related Article algorithm, a keyword-based relatedness measure developed by the National Library of Medicine.
- They then collect all patents that cite these related publications.
- This measure is designed to capture the *net* effect of NIH funding on total private patenting in an area, including indirect channels and allowing for crowd-out.

2.6 Fractional versus unit patent counts

- A patent can be linked to many DSTs.
- **Fractional counts:** if a patent links to N DSTs, each DST gets $1/N$ of the patent. This is natural if knowledge inputs are substitutes.
- **Unit counts:** each linked DST gets a full patent. This is natural only if all linked knowledge inputs are essential complements.
- For broad “related patent” measures, the paper prefers the conservative **fractional-count** interpretation because indirect matching often links patents to many study sections.

2.7 Descriptive statistics that matter

- **Grant to patent linkage.** 66,085 grants, about 43% of the grant sample, produce a publication directly cited by a patent.
- **Direct citation prevalence.** 44% of all life-science patents and 39% of private-sector patents directly cite NIH-funded research.
- **Broader relevance prevalence.** About 84% of all life-science patents and 82.3% of private-sector patents are related to NIH-funded research through the broader related-publication mapping.
- **High-value patents are rare.** Only 4,718 private-sector patents are linked to advanced drug candidates, and only 1,999 are linked to FDA-approved drugs.
- **Average DST size.** The average DST supports 10.85 grants and about \$40.6 million of funding.
- **Average patent output per DST.** The average DST is linked to 12.82 fractional citation-linked private patents and 24.84 fractional related private patents.

2.8 Lag and spillover patterns in the raw data

- The grant-to-patent lag is long and variable: the cumulative probability that a grant eventually links to a patent rises for many years after the grant is awarded.
- Roughly fifteen years after funding, more than 60% of grants funded by the National Institute of Allergy and Infectious Diseases have produced research cited by patents, versus only about 20% for the National Institute of Mental Health.
- Time-to-patent shortens across grant cohorts: for grants awarded in 1980–1985, only about 20% are linked to a patent within ten years; for grants awarded in 1991–1995, the ten-year figure is close to 40%.
- Important drug patents are rare: even after twenty years, fewer than 5% of grants produce a patent associated with an FDA-approved drug.
- Same-disease and different-disease patent linkages are very similar in the lag graphs, foreshadowing the paper’s later finding that over half of induced patents spill across disease areas.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual innovation production function

The paper’s conceptual object is an innovation production function in which patenting in one area can depend on knowledge produced in many other areas and at many different times:

$$p_{\nu\tau} = f(\{k_{rt}\}_{r,t}). \quad (1)$$

- **Interpretation.** Patent output in research area ν at time τ can depend on knowledge generated in many research areas r and many dates t .
- **Why it matters.** Earlier work often imposes $r = \nu$ and fixed lags. This paper tries to avoid both restrictions by using bibliometric linkages.

3.2 Generic empirical object

The authors estimate reduced-form relationships between funding for a DST and later private patenting outcomes linked to that DST:

$$Patents_{dst} = \alpha_0 + \alpha_1 Funding_{dst} + Controls_{dst} + \varepsilon_{dst}. \quad (3.1)$$

- **Dependent variable.** $Patents_{dst}$ is alternatively the number of citation-linked private patents or the number of related private patents.
- **Regressor of interest.** $Funding_{dst}$ is total NIH funding in disease d , science area s , and year t .
- **Interpretation of α_1 .** The coefficient measures how much extra private patenting is associated with an extra dollar of NIH funding for that research area.

3.3 Main fixed-effects specification

$$Patents_{dst} = \alpha_0 + \alpha_1 Funding_{dst} + \beta' X_{dst} + \delta_{ds} + \gamma_{dt} + \nu_{st} + \varepsilon_{dst}. \quad (3.2)$$

- δ_{ds} . Disease-science fixed effects absorb time-invariant differences in commercial and scientific potential across research areas.
- γ_{dt} . Disease-year fixed effects absorb disease-specific shocks over time, such as shifts in disease burden or political salience.
- ν_{st} . Science-year fixed effects absorb shocks to scientific opportunity that hit a methodology or scientific field across diseases.
- **Additional controls.** In the most saturated specification the authors also include fixed effects for the number of applications a DST receives, as a proxy for time-varying interest in the area.
- **Weights and inference.** Regressions are weighted by average DST size and standard errors are double-clustered by disease and science.
- **Identification after fixed effects.** The coefficient is identified from within-disease-year and within-science-year funding differences across disease-science pairings.

3.4 Instrumented specification

Because even residual within-cell funding variation may reflect better peer-review scores for “hot” areas, the paper estimates an IV version with richer first-stage controls:

$$Patents_{dst} = \alpha_0 + \alpha_1 Funding_{dst} + \Upsilon(\#Applications_{dst}) + \Phi(RawScores_{dst}) + \Psi(ScienceRanks_{dst}) + \delta_{ds} + \gamma_{dt} + \nu_{st} + \varepsilon_{dst} \quad (3.3)$$

- **Application-count controls.** These absorb variation coming simply from having more applications near the cutoff.
- **Raw-score and science-rank controls.** These flexibly partial out grant quality as assessed by reviewers inside the funding window.
- **Interpretation.** After conditioning on these objects, remaining variation comes from the way within-study-section science ranks are translated into Institute-level funding priority, not from obvious quality differences.

3.5 Windfall funding instrument

$$WindfallFunding_{dst} = Funding_{dst}^{W_{dt}} - \mathbb{E}\left[Funding_{dst}^{W_{dt}}\right]. \quad (3.4)$$

- W_{dt} . The set of applications within a narrow window around disease area d ’s payline in year t .
- **Baseline bandwidth.** In the main specification, W_{dt} is the set of **25 applications on either side** of the Institute-specific funding threshold.
- **Expected funding.** If funding were random within this window, a DST would expect half its applications in the window to be funded. Expected funding is therefore constructed from the number of marginal applications times one-half times the mean funded grant size in that disease-year.

- **Economic idea.** A positive windfall means the DST had more marginal applications funded than expected; a negative windfall means fewer.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The fixed-effects design compares private patenting across disease-science research areas that receive different amounts of funding within the same disease-year and within the same science-year.
- The idea is to hold fixed broad disease shocks and broad scientific-opportunity shocks, then ask whether extra funding to one disease-science pairing predicts more private follow-on innovation.
- The main concern is that the NIH may still funnel money toward areas with higher latent commercial potential.

4.2 Why the NIH’s institutional design helps

- Study sections review grants by *science*, while Institutes fund them by *disease*.
- Because Institutes must fund applications in order of science rank until the budget is exhausted, they cannot easily cherry-pick all the best grants in one disease-science niche without also funding all better-ranked grants from other science niches in the same disease area.
- This mechanical “rank of ranks” system creates quasi-random differences in realized funding across DSTs near the payline.

4.3 How the windfall IV works

- Consider all grant applications in a narrow window around an Institute’s payline.
- If funding is random within that window, a DST with six applications there should expect three to be funded.
- If four end up funded, the DST receives a positive funding windfall; if two are funded, it receives a negative one.
- The stylized example in Figure 2 shows exactly this logic with the cancer/cell-signalling and cancer/tumor-physiology DSTs.

4.4 Why the exclusion restriction looks plausible in the paper

- The paper conditions flexibly on the **number** of applications near the cutoff, the **average raw score**, and the **average science rank** of those applications.
- After this conditioning, variation in windfall funding comes from the fact that identical within-study-section scores can translate into different Institute-level funding outcomes once applications from different study sections are pooled inside the disease budget.

- Table 4 shows that funding priority within the 25-grant window does *not* predict publications, Bayh-Dole patents, citation-linked private patents, advanced-drug-candidate patents, or FDA-approved-drug patents.
- Table 5 and Figure 8 show that current windfalls are strongly related to current funding, but not to past windfalls, future windfalls, or non-windfall funding.

4.5 First-stage strength

- The first-stage coefficient of windfall funding on total DST funding is **1.098**.
- The paper reports a Cragg–Donald Wald F -statistic of **344** and a Kleibergen–Paap Wald F -statistic of **24.48**, both supportive of a non-weak instrument.

4.6 What OLS and IV each identify

- **OLS fixed effects** capture both anticipated and unanticipated funding changes.
- **IV** isolates unanticipated funding shocks from marginal grant outcomes around the payline.
- The authors view the two approaches as complementary: OLS is broader but more vulnerable to endogenous targeting; IV is cleaner but identifies a narrower source of variation.

5 Tables and figures

5.1 Figure 1: Overview of the bibliometric linkage strategy

Read:

- The figure is the paper’s roadmap. It starts with NIH administrative grant data, links grants to PubMed publications through acknowledgments, then links publications to patents through patent citations.
- The right branch adds a publication-to-publication relatedness step, which is how the paper builds the broader “related patent” measure.
- The big conceptual payoff is that the authors do not impose a disease-specific lag structure. The data themselves reveal where the downstream commercial impact shows up.

5.2 Figure 2: Example of windfall DST funding

Read:

- The example compares two cancer DSTs: cancer/cell signalling and cancer/tumor physiology.
- Inside the 5-grant window around the payline, both DSTs have the same number of applications and the same mean raw score and mean science rank, yet one gets a \$1 million negative windfall and the other a \$1 million positive windfall.
- This is the core intuition for the IV: conditional on quality and the number of marginal applications, realized funding differs because of procedural funding mechanics rather than obvious scientific merit differences.

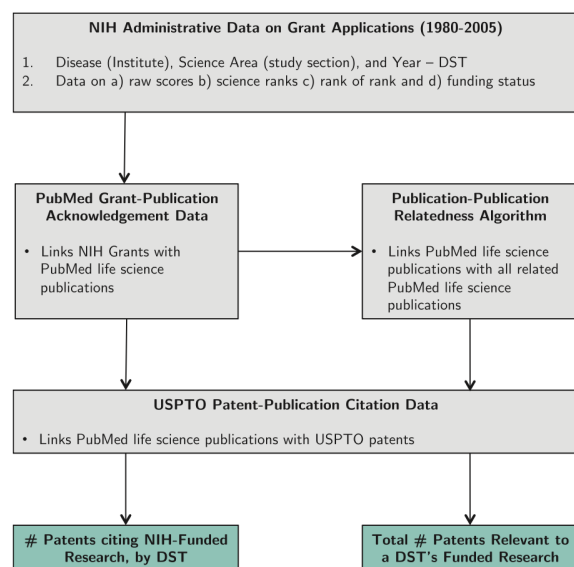


FIGURE 1

Overview of data and construction of patent outcome measures.

Note: Please consider this figure after References.

Cell Signaling Study Section				Tumor Physiology Study Section			
Grant ID	Disease	Raw Score	Science Rank	Grant ID	Disease	Raw Score	Science Rank
G1	Cancer	1.0	1	G16	Other	1.1	1
G2	Other	1.1	2	G17	Other	1.2	2
G3	Other	1.2	3	G18	Cancer	1.3	3
G4	Cancer	1.3	4	G19	Other	1.4	4
G5	Cancer	1.4	5	G20	Cancer	1.5	5
G6	Other	1.6	6	G21	Cancer	1.6	6
G7	Cancer	1.7	7	G22	Cancer	2.1	7
G8	Cancer	2.4	8	G23	Other	2.2	8
G9	Other	2.5	9	G24	Cancer	2.3	9
G10	Other	2.8	10	G25	Cancer	2.8	10
G11	Other	2.9	11	G26	Other	2.9	11
G12	Cancer	3.2	12	G27	Other	3.1	12
G13	Cancer	3.4	13	G28	Other	3.3	13
G14	Other	3.6	14	G29	Cancer	3.5	14
G15	Other	3.7	15	G30	Cancer	3.6	15
				G31	Cancer	3.7	16

Cancer Institute (NCI)				Comparison of Outcomes (assuming each grant requests \$2M)	
Grant ID	Study Section	Raw Score	Science Rank	Funding Priority	
G1	Cell	1.0	1	1	
G18	Tumor	1.3	3	2	
G4	Cell	1.3	4	3	
G5	Cell	1.4	5	4	
G20	Tumor	1.5	5	5	
G21	Tumor	1.6	6	6	
G7	Cell	1.7	7	7	
G22	Tumor	2.1	7	8	
G8	Cell	2.4	8	9	
G24	Tumor	2.3	9	10	
G25	Tumor	2.8	10	11	
G12	Cell	3.2	12	12	
G13	Cell	3.5	13	13	
G29	Tumor	3.5	14	14	
G30	Tumor	3.6	15	15	
G15	Cell	3.7	15	16	
G31	Tumor	3.7	16	17	

Cancer CS			Cancer TP
All applications			
# of Apps	8	9	
Mean Raw Score	2.30	2.5	
Mean Science Rank	8.13	9.44	
Total DST Funding	\$10M	\$12M	
In 5-grant window			
# of Apps in Window	5	5	
Mean Raw Score	2.9	2.9	
Mean Science Rank	11	11	
Windfall DST Funding	-\$1M	\$1M	

FIGURE 2

Example of windfall DST funding.

Notes: This figure illustrates an example of our windfall funding instrument. Please see the text in Section 3.2.2 for details. Grants are funded by NIH Institutes in order of their science rank, using raw scores as tiebreakers.

5.3 Table 1: Grant characteristics, 1980–2005

Read:

- This table fixes the scale of the underlying grant data: 153,076 grants, 17 disease areas, 624 study sections, and 14,085 DSTs.
- The average grant is about \$1.56 million, 74% are R01-equivalent, and about 60% are new projects.
- Grants that eventually link to patents are not random: they tend to have more publications and larger funding amounts, which is why the paper later puts so much emphasis on identification.

5.4 Table 2: Patent characteristics, 1980–2012

Read:

- This table shows that NIH-funded science is deeply embedded in private patenting: 39.4% of private-sector patents directly cite NIH-funded research and 82.3% are related to NIH-funded research under the broader metric.
- High-value patents are rare in levels, but especially NIH-connected in shares: about half of advanced drug-candidate patents directly cite NIH-funded research.
- Small assignees are more likely than large assignees to cite or relate to NIH-funded science, which anticipates the later heterogeneity results.

5.5 Figures 3 and 4: Grant-patent lag structure by disease area and cohort

Read:

TABLE 1
Grant characteristics, 1980–2005

	Full sample	Grants linked to private-sector patents	
		Cited by patents	Related to patents
Sample coverage			
No. of Grants	153,076	66,085	123,872
No. of Disease areas (institutes)	17	17	17
No. of Science areas (study sections)	624	548	598
No. of DSTs	14,085	9,951	13,092
Grant characteristics			
Per cent of R01 equivalent grants	73.74	77.46	74.33
Per cent of Centre Grants	3.26	4.79	3.20
Per cent of Teaching or fellowship grants	11.43	10.12	11.27
Per cent of New	59.50	51.08	58.55
Funding amount (total project allocation, 2010 dollars; mean & SD)	\$1,556,969 (2,198,506)	\$1,875,779 (2,783,272)	\$1,568,881 (2,215,371)
No. off acknowledged publications	1.41 (3.58)	3.27 (4.86)	1.75 (3.91)
No. of related publications	84.80 (194.36)	166.10 (271.34)	104.90 (211.24)
No. of patents citing grant (weighted counts)	0.43 (2.36)	1.00 (3.51)	0.54 (2.62)
No. of patents related to grant (weighted counts)	0.84 (2.21)	1.60 (3.05)	1.04 (2.41)

Notes: Sample is the set of all NIH-funded grants from 1980 to 2005, excluding NINR, NLM, and NIMHD grants (see Appendix A for a full list of ICs in the sample) and evaluated by chartered study sections. The sample is restricted to new and competitive renewal grants so that there is one observation per successful grant application cycle. A grant is defined as cited by patents if there exists a patent that cites a publication that acknowledges funding from that grant. A grant is matched with a publication if it acknowledges the project number of the grant and is published within 5 years of the grant's funding year. A patent is citation-linked to a grant if it cites a publication that is linked to a grant. A grant is considered related to a patent if that grant produces a publication that is similar (as defined by the PubMed Relatedness Matching Algorithm) to a publication that is cited by a patent. In this article, we require that similar publications be published within 5 years of each other. A grant is an R01 equivalent (e.g. a large project-based grant) if its NIH funding mechanism is either an R01, R23, R29, or R37. Centre grants are those grants whose mechanism starts with a "P" (e.g. a P01 grant containing multiple projects). A teaching or fellowship grant is one whose grant mechanism designation begins with a "T" or an "F". New grants are projects that have not previously received NIH funding. Acknowledged publications are the unique count of PubMed publications which acknowledge the grant's main project number and which are published within 5 years of grant receipt. Related publications include directly acknowledged publications, in addition to all publications related to them, according to the PMRA algorithm discussed in the text, and published within a 5 year window.

TABLE 2
Patent characteristics, 1980–2012

	Full sample	Patents linked to NIH-funding	
		% Citing NIH funded research	% Related to NIH funded research
Sample coverage			
No. of patents	315,982	44.00	84.10
Patent characteristics:	47245	105248	
general			
Private sector	232,276	39.38	82.33
Public sector	83,394	56.91	89.07
Patent characteristics:			
private sector only			
Advanced drug candidates	4,718	49.92	88.22
FDA-approved drugs	1,999	42.47	86.79
Large assignee	164,431	36.23	80.37
Small assignee	29,183	51.37	87.89

Notes: Sample is the set of all USPTO granted patents from 1980 to 2012 that meet the following criteria: (1) they are either in NBER Patent Categories 1 ("Chemicals") or 3 ("Drugs and Medical") and (2) they cite at least one publication in the PubMed database. A patent is defined as citing NIH-funded research if it cites a publication that acknowledges the project number of an NIH grant and is published within 5 years of that grant's funding year. A patent is considered related to NIH funding if it cites a publication that is similar (as defined by the PubMed Relatedness Matching Algorithm) to a publication that acknowledges NIH funding. We require that similar publications be published within 5 years of each other. A patent is labelled "Private Sector" if it is assigned to a domestic U.S. or foreign corporation (NBER assignee categories 1 and 2 minus foundations, universities, and hospitals). A patent is labelled "Public Sector" if it is assigned to a U.S. or foreign government (NBER categories 5 and 6) or if it is assigned to a foundation, university, or hospital. A patent is labelled an advanced drug candidate if it is associated with a drug or biologic in Phase III clinical trials or beyond (these are listed in Orange Book and/or IMS Patent Focus); a patent is associated with an FDA-approved drug if that patent is associated with a marketed treatment according to IMS Health. A patent is associated with a large assignee if its assignee employs over 500 employees; it is considered small otherwise.

- Figure 3 shows large heterogeneity in how fast and how often grants translate into patents across disease areas. At around fifteen years, some Institutes exceed 60% while others remain near 20%.
- Figure 4 shows that the translation from grant to patent becomes faster over later grant cohorts. Later cohorts reach a given cumulative patent-linkage rate much sooner.
- These figures justify the paper's decision not to hard-wire one disease-specific lag into the empirical design.

5.6 Figures 5 and 6: Lag structure by patent quality and disease spillovers

Read:

- Figure 5 makes the point that patent linkage is common, but *commercially important* patent linkage is rare. Even after many years, only a small fraction of grants map to FDA-approved-drug patents.
- Figure 6 shows that same-disease and different-disease curves are extremely close, suggesting large cross-disease spillovers right in the raw data.
- The figure also shows that both private and public sectors draw on NIH-funded science.

5.7 Table 3 and Figure 7: Outcome distributions at the DST level

Read:

- Table 3 shows the average DST supports 10.85 grants and \$40.6 million of funding, with 12.82 fractional citation-linked patents and 24.84 fractional related patents.
- Figure 7 shows these outcomes are highly skewed. Most DSTs produce relatively few patents, while a small number sit far out in the right tail.
- This is why weighting by average DST size matters: without it, tiny DSTs would have too much leverage.

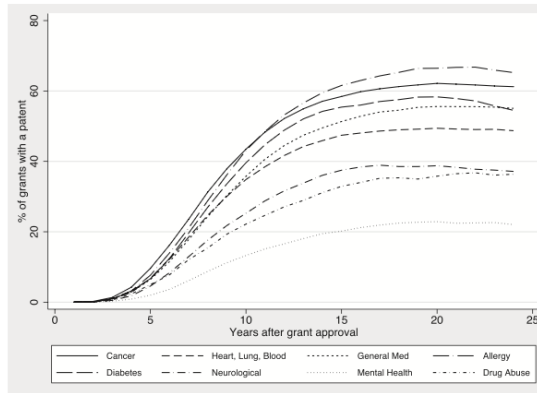


FIGURE 3
Grant-patent lags by disease area—top 10 ICs.

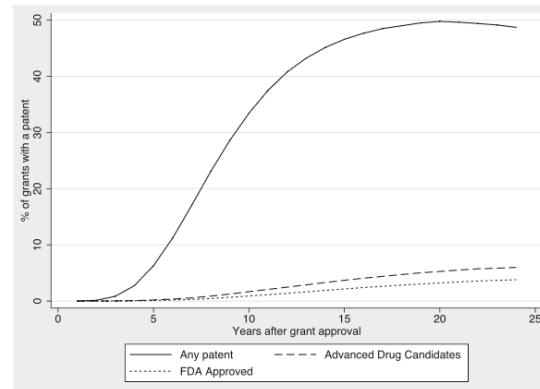


FIGURE 5
Grant-patent lags by patent quality.

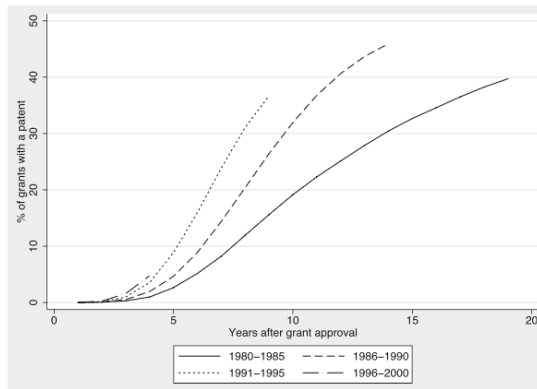


FIGURE 4
Grant-patent lags by grant cohort.

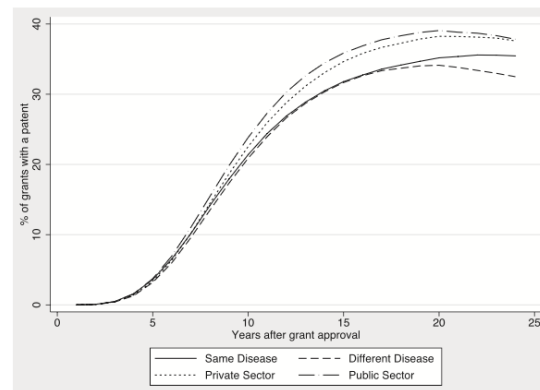


FIGURE 6
Grant-patent lags by patent type.

TABLE 3
NIH research area (DST) characteristics, 1980–2005

	Full sample	DSTs linked to patents	
		Cited by patents	Related to patents
Average no. of grants	10.85 (16.58)	15.60 (19.05)	11.62 (17.01)
Output characteristics			
Funding amount (DST)	\$40,631,460 (43,611,800)	\$45,556,350 (44,448,260)	\$41,397,230 (43,683,690)
No. of patents citing NIH-funded research (fractional counts)	12.82 (19.17)	14.71 (19.85)	13.07 (19.28)
No. of patents citing NIH-funded research (unit counts)	101.7 (153.6)	116.8 (159.1)	103.7 (154.4)
No. of patents related to NIH-funded research (fractional counts)	24.84 (27.95)	28.33 (28.31)	25.30 (28.00)
No. of patents related to NIH-funded research (unit counts)	3,520 (3,742)	4,023 (3,755)	3,589 (3,745)
N	14,085	8,886	13,027

Notes: Sample is the same as that in Table 1, except aggregated to the NIH Disease/Science/Time level. See the notes to Table 1 for additional definitions. The funding and patent variables are weighted by average DST size, that is, the average yearly number of grants in a Disease/Science research area. In fractional patent counts, a patent matched to N distinct DSTs counts as $1/N^{\text{th}}$ of a patent for each DST. In unit patent counts, a single patent matched to N distinct DSTs counts as one patent for each DST. Funding amounts are expressed in 2010 dollars (deflated by the Biomedical R&D Producer Price Index).

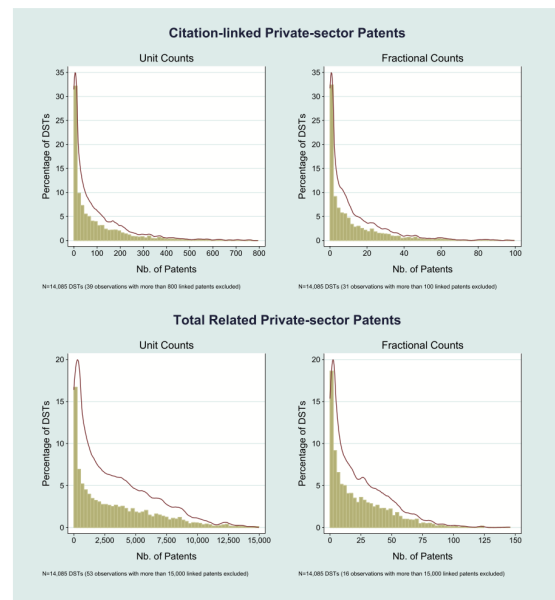


FIGURE 7
Outcome measures by DST.

5.8 Table 4: Funding priority and grant outcomes around the payline

Read:

- This is the paper’s most direct “near-cutoff quality” test.
- Within the 25-grant window, better funding priority does *not* predict more publications, more Bayh-Dole patents, more citation-linked private patents, more advanced-drug-candidate patents, or more FDA-approved-drug patents.
- The coefficients are tiny and statistically insignificant, which supports the claim that near the payline funding outcomes are conditionally close to random.

TABLE 4
Relationship between funding priority and grant outcomes

	Direct acknowledgments		Citation-based patent linkages		
	Publications (1)	Patents (2)	All private sector (3)	Adv. Drug candidates (4)	FDA approved Drugs (5)
Funding priority	-0.007 (0.004)	0.085 (0.172)	0.368 (0.356)	0.020 (0.036)	0.010 (0.027)
R ²	0.483	0.293	0.376	0.222	0.198
Observations	8,704	8,704	8,704	8,704	8,704
Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Science FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Science × Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.

Notes: Each observation is a single funded grant that falls within a 25-grant distance from an NIH Institute’s payline. Publications are defined as publications acknowledging the grant, published within 5 years of the grant’s receipt. Patents are patents that acknowledge funding by that grant, as mandated by the Bayh-Dole Act. Citation-based patent linkages are defined as a patent citing a publication that in turn acknowledges funding from that grant. Funding priority is the grant’s ordering for that disease-year. Standard errors in parentheses, two-way clustered at the disease and science level (*p < 0.10, **p < 0.05, ***p < 0.01).

TABLE 5
Alternative first stages, past and future windfalls

	Dependent variable: total DST funding		
	Past windfall (1)	Current windfall (2)	Future windfall (3)
Windfall funding	-0.078 (0.203)	1.098*** (0.209)	0.148 (0.171)
R ²	0.919	0.926	0.916
Observations	9,326	14,085	9,326

Notes: This table presents alternative first stages using past and future windfall funding. Current windfall funding is total funding (2010 dollars) minus expected funding, for the subset of grants funded by a DST whose rank of rank scores were marginal, that, were within 25 applications of the award cut-off for their specific disease area (Institute). To calculate expected funding, we multiply the total number of DST applications within this window by 1/2 and by the mean funding amount for approved grants in that disease-year. Future windfall is this same amount, but defined for DST+1. Past windfall funding is similarly defined, for DST-1. Controls include disease-science and disease-year fixed effects, linear science-year time trends, as well as fixed effects for the number of applicants to a DST, the number of applicants within a 25-grant radius window around the IC payline, as well as cubics in the average raw and rank scores of applications in the funding window. The outcome variables are fractional patent counts. Standard errors in parentheses, two-way clustered at the disease and science level (*p < 0.10, **p < 0.05, ***p < 0.01).

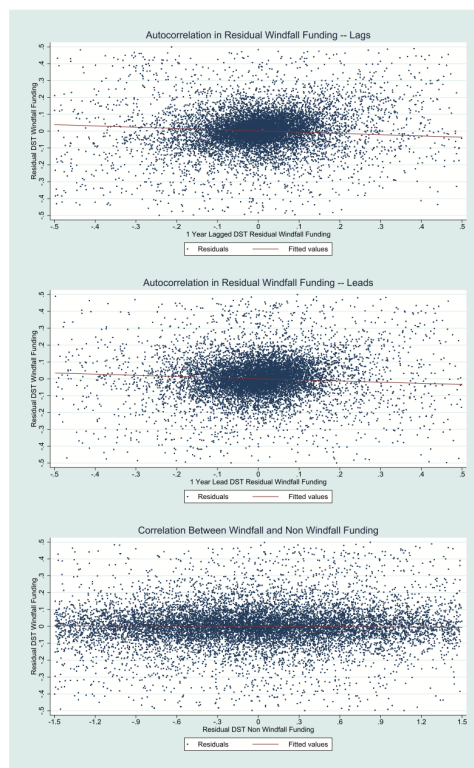


FIGURE 8
Correlation between windfall DST funding and other DST funding.

5.9 Table 5 and Figure 8: Placebo first stages and residual correlations

Read:

- Table 5 shows that current windfall funding strongly predicts current total funding (coefficient 1.098), but past and future windfalls do not.
- Figure 8 adds the same message graphically: the residual scatterplots for lagged windfalls, lead windfalls, and non-windfall funding are essentially flat.
- Together, these exhibit-level checks support the IV exclusion restriction.

5.10 Table 6: Effect of NIH funding on citation-linked private patents

Read:

- The core OLS result is very stable across specifications: with the full set of fixed effects and application-count controls, \$10 million of funding raises citation-linked private patents by about **2.45** in fractional counts.
- The corresponding unit-count estimate is 18.41, but the elasticities are very similar across the two counting schemes.
- The stability across columns is important: adding more demanding fixed effects does not wipe out the positive relationship.

	No. of patents citing NIH-funded research				
	(1)	(2)	(3)	(4)	(5)
Fractional patent counts: Mean=12.82; SD=19.17					
DST Funding (× \$10 mln.)	2.595***	2.281***	2.242***	2.550***	2.450***
Mean=4.06; SD=4.36					
Elasticity	(0.220)	(0.356)	(0.359)	(0.654)	(0.568)
R ²	0.822	0.723	0.71	0.808	0.777
	0.417	0.600	0.641	0.918	0.933
Unit patent counts: Mean=101.7; SD=153.6					
DST Funding (× \$10 mln.)	21.830***	17.831***	17.842***	18.626***	18.412***
Mean=4.06; SD=4.36					
Elasticity	(1.731)	(2.068)	(2.067)	(4.308)	(3.648)
R ²	0.872	0.712	0.713	0.744	0.735
Observations	14,085	14,085	14,085	14,085	14,085
Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Science FEs					
Disease × Year FEs					
Science × Year FEs					
Application count FEs					

Notes: Each observation is Disease/Science/Time (DST) combination. A patent is citation-linked to a DST if it cites research that acknowledges funding from that DST. For more details on this sample, see the notes to Tables 1 and 3. Funding is defined by the sum of project-cycle allocations for all new and competing renewal grants that are associated with that DST. The patent sample is restricted to those with private sector assignees, and weighted by average DST size, that is, the average yearly number of grants in a Disease/Science research area. See Table 2 for more details. Year FEs are fixed effects for the fiscal year associated with a DST. NIH Institutes are taken to represent diseases and NIH study sections (review committees) are taken to represent science areas. Elasticities are evaluated at sample means. Application count FEs are indicator variables for the number of applications that a DST receives. Standard errors in parentheses, two-way clustered at the disease and science level (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

	No. of patents related to NIH-funded research				
	(1)	(2)	(3)	(4)	(5)
Fractional patent counts: Mean=24.8; SD=28.0					
DST Funding (× \$10 mln.)	4.516***	3.593***	3.590***	3.712***	3.239***
Mean=4.06; SD=4.36					
Elasticity	(0.210)	(0.512)	(0.537)	(0.601)	(0.372)
R ²	.738	0.588	0.587	0.607	0.530
	0.536	0.759	0.783	0.965	0.974
Unit Patent Counts: Mean=3,969; SD=3,918					
DST Funding (× \$10 mln.)	603.082***	456.685***	453.133***	504.728***	445.983***
Mean=4.06; SD=4.36					
Elasticity	(26.714)	(53.002)	(56.424)	(80.237)	(41.404)
R ²	0.696	0.527	0.523	0.583	0.515
Observations	14,085	14,085	14,085	14,085	14,085
Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Science FEs					
Disease × Year FEs					
Science × Year FEs					
Application Count FEs					

Notes: Each observation is Disease/Science/Time (DST) combination. A patent is considered to be in the same area as an NIH grant if it cites a publication that is similar (as defined by the PubMed Relatedness Matching Algorithm) to a publication that is linked to a patent. For more details on this sample, see the notes to Tables 1 and 2. Funding is defined by the sum of project-cycle allocations for all new and competing renewal grants that are associated with that DST. The patent sample is restricted to those with private sector assignees, and weighted by average DST size, that is, the average yearly number of grants in a Disease/Science research area. See Table 2 for more details. Year FEs are fixed effects for the fiscal year associated with a DST. NIH Institutes are taken to represent diseases and NIH study sections (review committees) are taken to represent science areas. Elasticities are evaluated at sample means. Application count FEs are indicator variables for the number of applications that a DST receives. Standard errors in parentheses, two-way clustered at the disease and science level (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

5.11 Table 7: Effect of NIH funding on total related private patents

Read:

- This table asks the stronger crowd-out question: does more NIH funding raise *total* private patenting in the affected intellectual area?
- In the preferred fractional-count specification, the full-control coefficient is about **3.24** patents per \$10 million.
- Because the effect on total related patenting is positive and at least as large as the direct citation-linked effect, the paper does not find evidence consistent with full crowd-out.
- The huge unit-count estimates are not the preferred interpretation because the related-patent mapping links patents to many DSTs, so treating every link as pivotal is too strong.

5.12 Table 8: IV estimates using windfall funding

Read:

- The first stage is strong, and the IV estimates remain positive.
- For citation-linked patents, the OLS and IV estimates are similar: about 2.41 versus 2.27 patents per \$10 million.

- For total related patents, OLS gives 3.63 and IV gives **2.67**; the authors take the IV number as their preferred estimate.
- This is the paper’s main causal punchline: even after using quasi-random funding shocks, public biomedical research spending increases private-sector patenting.

TABLE 8
Effect of NIH investments on private-sector patenting windfall funding IV

	First stage DST funding (× \$10 mln.)	Citation linked Mean=2.82; SD=9.17		Total related Mean=24.8; SD=28.0	
		OLS (2)	IV (3)	OLS (4)	IV (5)
Windfall funding (×\$10 mln.)	1.098*** (0.209)	DST Funding (×\$10 mln.) Mean=4.06; SD=4.36 Elasticity 0.763	2.408*** (1.228) 0.720	3.625*** (0.807) 0.593	2.668* (1.368) 0.437
Cragg-Donald Wald F-stat	344				
Kleibergen-Paap Wald F-stat	24.48				
Observations	14,085	14,085	14,085	14,085	14,085
Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Science FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Disease × Year FEs	Incl.	Incl.	Incl.	Incl.	Incl.
Science × Year Linear	Incl.	Incl.	Incl.	Incl.	Incl.
Trends					
Application controls	Incl.	Incl.	Incl.	Incl.	Incl.

Notes: See notes to Tables 6 and 7 for details about the sample. The outcome variables are fractional patent counts. The instrument is total funding (2010 dollars) minus expected funding, for the subset of grants funded by a DST whose rank of rank scores were marginal, that is, were within 25 applications of the award cut-off for their specific disease area (Institute). To calculate expected funding, we multiply the total number of DST applications within this window by 1/2 and by the mean funding amount for approved grants in that disease-year. Application controls include (1) FEs for the number of applications that a DST receives; (2) FEs for the number of applications associated with a DST that are also in a 50-grant window around the relevant IC pipeline, as well as (3) cubics in the average raw and rank scores of applications associated with a DST that are also in a 50-grant window around the pipeline. Elasticities are evaluated at the sample means. Standard errors in parentheses, two-way clustered at the disease and science level (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

TABLE 9
Effect of NIH investments on private-sector patenting heterogeneity by patent type

	All private sector	Advanced drug candidates	Highly cited	Same area	Different area	Large assignee	Small assignee
	Mean=24.8; SD=28.0 (1)	Mean=0.546; SD=0.864 (2)	Mean=1.28; SD=1.76 (3)	Mean=18.9; SD=23.8 (4)	Mean=15.9; SD=19.0 (5)	Mean=17.5; SD=20.7 (6)	Mean=3.47; SD=4.18 (7)
OLS							
DST funding (× \$10 mln.)	3.625*** (0.807)	0.080*** (0.015)	0.178*** (0.050)	2.686*** (0.447)	2.312*** (0.702)	2.562*** (0.614)	0.508*** (0.097)
Mean=4.06; SD=4.36 Elasticity	0.593	0.595	0.565	0.577	0.590	0.594	0.594
IV							
DST funding (× \$10 mln.)	2.668* (1.368)	0.078* (0.045)	0.156** (0.077)	1.421* (0.861)	2.221** (1.075)	1.988** (0.907)	0.534** (0.241)
Mean=4.06; SD=4.36 Elasticity	0.437	0.580	0.495	0.305	0.567	0.461	0.625
Observations	14,085	14,085	14,085	14,085	14,085	14,085	14,085

Notes: See notes to Tables 6 and 7 for sample details. The outcome variables are fractional patent counts. All specifications include disease-science FEs, disease-year FEs, science by year linear time trends, FEs for the number of applications to the DST, cubics in the average raw score and average science rank received by applications in the 25-grant radius window around the IC pipeline, and FEs for number of DST applicants in a 25-grant window around an IC’s funding cut-off. A patent is labelled “Private Sector” if it is assigned to a domestic US or foreign corporation (NBER assignee categories 1 and 2 minus foundations, universities, and hospitals). A patent is labelled an advanced drug candidate if it is included in IMS Patent Focus, which has information on patents on drugs in Phase III trials or further. A patent is in the same disease area as a DST if the majority of NIH research areas that it is linked are also associated with that same “D” disease area. A patent is associated with a large assignee if its first assignee employs more than 500 employees; it is considered small otherwise. Standard errors in parentheses, two-way clustered at the disease and science level (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

5.13 Table 9: Heterogeneity by patent type

Read:

- Column 2 shows that \$10 million of funding raises advanced-drug-candidate patents by about **0.078** in IV, with an elasticity around 0.58.
- Column 3 shows a positive effect on highly cited patents, so the paper is not only generating low-value patent counts.
- Columns 4 and 5 show the cross-disease result directly: same-disease patents rise by about **1.42**, while different-disease patents rise by about **2.22**. More than half the induced innovation spills across disease boundaries.
- Columns 6 and 7 show positive effects for both large and small assignees. Large firms account for more patents in levels, but the elasticity for small firms is at least as large.

5.14 Table 10: Implied drug valuation

Read:

- The table converts the patent estimates into rough private-value numbers using drug sales data.
- The central IV estimate in Column 2 implies that \$10 million in NIH funding yields about **0.054** FDA-approved-drug patents. If a drug has about eight relevant patents on average, that maps into about **\$23.4 million** in downstream drug sales.
- Alternative assumptions using pre-approval patents, main patents, or drug-level outcomes give values roughly between **\$13.9 million** and **\$24.3 million**.
- The table is explicitly back-of-the-envelope, but it reinforces the paper’s conclusion that the private return is positive.

TABLE 10
Implied drug valuation of NIH investments

	Advanced drug candidates Mean=0.546; SD=0.864 (1)	FDA approved Mean=0.316; SD=0.532 (2)	Pre-approval Mean=0.212 SD=0.358 (3)	Main Mean=0.035; SD=0.084 (4)	Drug-level Mean=0.059; SD=0.099 (5)
OLS					
DST funding ($\times$ \$10 mln.) Mean=4.06; SD=4.36	0.080*** (0.015)	0.046*** (0.012)	0.032*** (0.007)	0.005*** (0.001)	0.008*** (0.002)
Elasticity	0.595	0.591	0.613	0.580	0.551
Implied drug value (\$ mln.)	—	\$20.0	\$22.2	\$17.4	\$27.8
IV					
DST Funding ($\times$ \$10 mln.) Mean=4.06; SD=4.36	0.078* (0.045)	0.054** (0.024)	0.033* (0.019)	0.004 (0.004)	0.007 (0.005)
Elasticity	0.580	0.694	0.632	0.464	0.482
Implied drug value (\$ mln.)	—	\$23.4	\$22.9	\$13.9	\$24.3
Observations	14,085	14,085	14,085	14,085	14,085

Notes: See notes to Tables 6 and 7 for sample details. The outcome variables are fractional patent counts. All specifications include disease-science FEs, disease-year FEs, science by year linear time trends, FEs for the number of applications to the DST, cubic in the average raw score and average science rank received by applications in the 25-grant radius window around the IC pipeline, and FEs for number of DST applicants in a 25-grant window around an IC's funding cutoff. A patent is labelled "Private Sector" if it is assigned to a domestic U.S. or foreign corporation (NBER assignee categories 1 and 2 minus foundations, universities, and hospitals). A patent is labeled an advanced drug candidate if it is included in IMS Patent Focus, which contains information on patents on biopharmaceutical candidates in Phase III trials or further. We do not generate an implied value for these patents since they are not necessarily associated with an approved drug/biologic. Within this set, patents are labeled as "FDA approved" if linked to an approved drug/biologic. A patent is labeled "pre-approval" if it is "FDA approved" and was filed prior to the time at which corresponding received marketing approval. A patent is labeled as "main" patent if it is the first patent ever filed associated with a marketed drug. Column 5 aggregates results to the drug level, reweighting by the number of unique drugs associated with a DST. Implied drug values are calculated assuming a mean lifetime discounted value of \$3.47 billion, in 2010 dollars. This figure comes from Dimasi *et al.*, 2004. All estimates assume that there is one pivotal patent per drug; FDA approved patents are scaled by 8; pre-approval patents by 5; main patents and drug specific outcomes are not scaled. For instance, the OLS estimate in column (2) implies that an additional \$10 mln. in NIH funding for a DST would result in \$20 mln. in downstream pharmaceutical sales. Standard errors in parentheses, two-way clustered at the disease and science level (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

6 Short summary

- **Conclusion.** The paper finds that NIH biomedical research funding causally raises private-sector patenting, even after addressing endogenous funding allocation and possible crowd-out.
- **Intuition.** Public science creates usable knowledge that private firms build on directly and indirectly, and the institutional rigidity of NIH funding rules creates quasi-random funding differences that let the authors measure this effect.
- **Empirical lesson.** Flexible grant-publication-patent linkages matter. If one only looked within the same disease area or imposed fixed lags, one would miss a large part of the effect.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** An additional \$10 million of NIH funding for a research area generates about **2.7** extra private-sector patents in related research areas in the preferred IV specification.
- **Measurement contribution.** The paper shows how to move from disease-level spending regressions to a much richer grant-publication-patent mapping that allows spillovers across diseases and long, uneven lags.
- **Crowd-out conclusion.** Because total related private patenting rises with NIH funding, the evidence is inconsistent with full crowd-out of private research effort.
- **Spillover conclusion.** More than half of induced patents occur outside the original disease area, which is central both economically and for policy design.
- **Value conclusion.** While patent valuation is noisy, nearly every calculation in the paper suggests a positive private return to public biomedical science spending.

7.2 Economic intuition

- Basic public science expands the stock of usable ideas that private firms can commercialize.
- Because biomedical knowledge travels across diseases and across long time horizons, the effect of a grant is much broader than its immediate disease label suggests.
- The NIH's funding system unintentionally creates useful research-design variation: study sections decide scientific rank, Institutes decide funding based on disease budgets, and this creates marginal funding shocks across disease-science cells.
- The positive results on both direct citation-linked patents and broader related patents suggest that public science is not merely substituting for private science; it is improving the technology opportunity set available to firms.

7.3 Takeaway

This paper is persuasive because it solves three problems at once: it measures the downstream private use of public science, it identifies quasi-exogenous variation in funding, and it checks whether public support merely displaces private innovation. The answer is that NIH funding does create private innovation, and it does so in ways that are broader than a simple disease-by-disease accounting would suggest. For anyone studying innovation policy, the paper is a benchmark example of how institutional detail and bibliometric linkage can be combined to estimate the spillover returns to public R&D.